

LFER Part 3: Curved Plots

Barry Linkletter

Hammett plots are not always simple. In rare cases they can be curved. When this happens we have an exciting and interesting observation to explain.¹

Class Discussion: Curved Hammett Plots

Be prepared to participate in the following exercises.

There are two ways a plot can be curved. If there are two different mechanisms for a reaction and they have different reaction constants, ρ , we may see one dominate and then the other as we change substituents.² If there is a mechanism with multiple steps and each step has a different dependence on electron density then we may see the rate-determining-step change as we change substituents. The rate is limited by the slowest step so we will see only the slower rate. Let us explore this idea.

1. Create two linear Hammett plots that cross (have the same rate) when we are using m-Cl as the substituent. Plot them both on the same graph.
 - (a) Add the two plots together over a range of σ values. This is analogous to two different mechanisms contributing to the overall rate. What does it look like?
 - (b) Only plot the line with the greater rate of the two over a range of σ values. This is analogous to the slower reaction being the r.d.s. What does it look like?
2. We have seen the hydrolysis of various forms of benzyl chloride and used Hammett plots to explore the mechanism. Let us now consider the solvolysis of benzyl tosylates in acetic acid.
 - (a) What is the tosylate group and why is it a common group in synthetic organic chemistry?
 - (b) Based on what you know about solvolysis of benzyl chloride in water and ethanol, propose a mechanism for this reaction.
 - (c) Table 1 presents rate data for a series of substituted benzyl tosylates.³ Plot this data as a Hammett plot. What do you observe?
 - (d) The mechanism is known to be S_N2 for benzyl tosylates in acetic acid.⁴ Given this fact propose an explanation for the Hammett plot.⁵

This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were created in ChemDoodle Plots were typeset using Matplotlib and Python tools. Diagrams were created and edited using Affinity Designer.

¹ Read chapter 8.3 and 8.4.

² This is similar to pH-rate profiles where multiple pathways are operating and we only see the largest contributor in each region of the plot. In a log-log plot we are exploring rate with orders of magnitude in the difference.

Figure 1: Acetolysis of benzyl tosylates

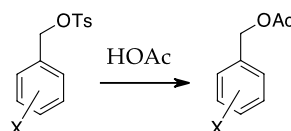


Table 1: Rates of solvolysis of benzyl tosylates in acetic acid at 40 °C.

Substituent	$k / 10^{-5} \text{ s}^{-1}$
p-CH ₃	96.3
m-CH ₃	4.54
p-F	4.19
H	1.71
p-Cl	0.86
m-Cl	0.087
m-CF ₃	0.0446
p-CF ₃	0.0277

Table 2: σ values for selected substituents.

Substituent	σ	σ^+	σ^-
p-CH ₃	-0.14	-0.31	-0.17
m-CH ₃	-0.06		
p-F	0.06	-0.07	-0.03
H	0		
p-Cl	0.23	0.11	0.19
m-Cl	0.37		
m-CF ₃	0.46		
p-CF ₃	0.53		0.65

³ "Acetolysis reactivities of substituted benzyl and polycyclic arylmethyl p-toluenesulfonates. Correlations with SCF- π and CNDO MO methods." A. Streitwieser Jr., et al., *J. Am. Chem. Soc.*, 1970, 92, 5141-5150 <https://doi.org/10.1021/ja00720a025>

⁴ Inversion is observed at the benzyl carbon in this reaction

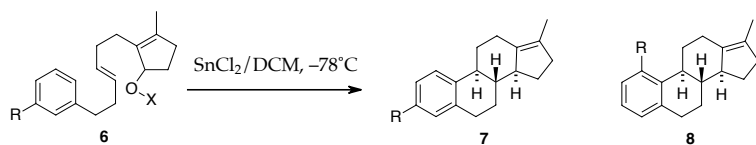
⁵ Ch. 11.5 discusses S_N1/S_N2 substitutions. Ch. 11.5.8 has a nice summary.

Investigating an Estrone Synthesis

An example of a synthetic plan⁶ that was investigated⁷ using the methods of physical organic chemistry. Please read the paper and the discussion below. Consider the novel step that defines the scheme and then examine the synthetic route to review your organic chemistry skills.

An Example of a Curved Hammett Plot

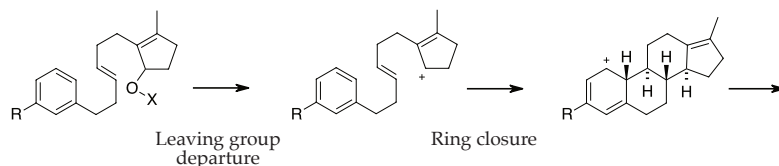
We begin with the mechanistic investigation. A long time ago in 1973, a synthesis of a steroid precursor, estrone, was reported.⁶ This synthesis was short, included reactions that you will be familiar with and involved a Hammett plot analysis to investigate the mechanism of the key step. The reaction that the authors were so proud of was the one-step multi-ring-closure reaction presented below.



⁶ "Stereospecific total synthesis of estrone via a cationic olefinic cyclization." P.A. Bartlett, W.S. Johnson, *J. Am. Chem. Soc.*, 1973, 95, 7501-7502. <http://doi.org/10.1021/ja00803a047>

Figure 2: The novel ring-closing reaction reported in this estrone synthesis route.

First we will consider the evidence for the proposed mechanism for this reaction as presented by the authors.⁷ The first consideration is the ordering of bond breaking and formation. The SnCl_2 Lewis acid activates the $-\text{O}-\text{X}$ group as a leaving group. Does it depart first and the electrophilic addition occurs in the following step (or steps) or is it a concerted process.



⁷ "Mechanism of a nonenzymic biogenetic-like olefinic cyclization.", P.A. Bartlett, J.I. Brauman, W.S. Johnson, R.A. Volkmann, *J. Am. Chem. Soc.*, 1973, 95, 7502-7504. <http://doi.org/10.1021/ja00803a048>

Figure 3: A possible stepwise mechanism for the formation of a cation followed by rate-determining ring closure.

In the step-wise mechanism presented in figure 3, the leaving group departs (assisted by the Lewis acid, not shown) to give a carbocation intermediate. Then an addition reaction (or a series of reactions) closes the rings and results in the electrophilic aromatic addition product shown. We know that the substituent on the benzene ring affects the rate, so the ring closure process must be the RDS. However, the authors report that the ratio of *para* to *ortho* products (7/8) is affected by the structure of the leaving group (see table 3.) If it has departed before the electrophilic aromatic addition, then it should not have any affect on the outcome. We conclude that the leaving group departs concurrently with aromatic addition and this backs up the idea of a single-step concerted process.

Table 3: *para/ortho* product ratios with changing structure of the leaving group, $-\text{OX}$

$-\text{R}$ group	$-\text{OX}$ group	7/8 ratio
$-\text{OTMS}$	$-\text{OH}$	8.4
$-\text{OTMS}$	$-\text{OTMS}$	2.6
$-\text{OTMS}$	$-\text{OBz}$	3.1
$-\text{OBz}$	$-\text{OH}$	2.4
$-\text{OBz}$	$-\text{OBz}$	1.3
$-\text{OCH}_3$	$-\text{OH}$	3.0

In the concerted mechanism presented in figure 4, the leaving group departs as the bonds are formed.

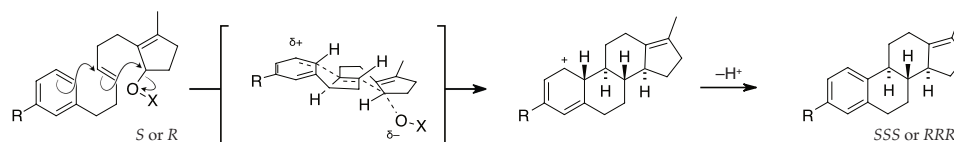


Figure 4: A possible concerted mechanism for ring-closing reaction.

This proposal is supported by the fact that the reaction is stereospecific in producing only *trans*-fused rings in the estrol backbone. The alkene must attack at the back side of the carbon with the leaving group. We know that elimination to create an alkene is *anti* in stereochemistry and so, according to the principle of microscopic reversibility, the addition must be *anti*. The all-*anti* transition-state is shown in figure 4. The original starting material had a racemic mixture for the stereochemistry at the leaving group and will therefore produce a racemic mixture for the product. But the relative stereochemistry in each enantiomer is identical.

We can conclude that the reaction is likely a concerted process. The fact that the leaving group affects the regiochemistry (*para/ortho* ratio, that no products are observed from potential cationic intermediates, and the stereospecificity of the cyclization support this idea.

The Experiment

The hypothesis can be stated and then tested.

"If the reaction is concerted and the benzene ring is involved in the r.d.s. then changing the electron density in the ring should affect the rate of the reaction. A Hammett plot should show a significant negative slope when a series of substituted benzene rings is used in the starting material."

So we need to synthesize a series of molecules with substituted benzene rings and plot the rate of the reactions *vs.* the substituent constant. The authors chose to use a hydroxyl group as the leaving group ($X = H$) and chose five substituents.⁸ The rate of reaction with SnCl_2 in DCM at -78°C was too fast to follow. The authors used ZnBr_2 as the Lewis acid in nitroethane at a temperature of -78°C and found that relative rates were able to be measured under these conditions.

The observation reported by the authors are present in table 4. They reported the rates as values relative to the rate observed for the trifluoromethyl case. I used those values to create rates relative to the unsubstituted case ($R = H$) and then produced the $\log(k_X/k_H)$ values for use in a Hammett plot. The substituent constants are presented in table 5. A plot of $\log(k_X/k_H)$ *vs.* σ is shown in figure 10.

The Hammett plot is curved. There is a region with a reaction constant of $\rho = -0.21$ and another with $\rho = -1.44$. The plot is "concave down",

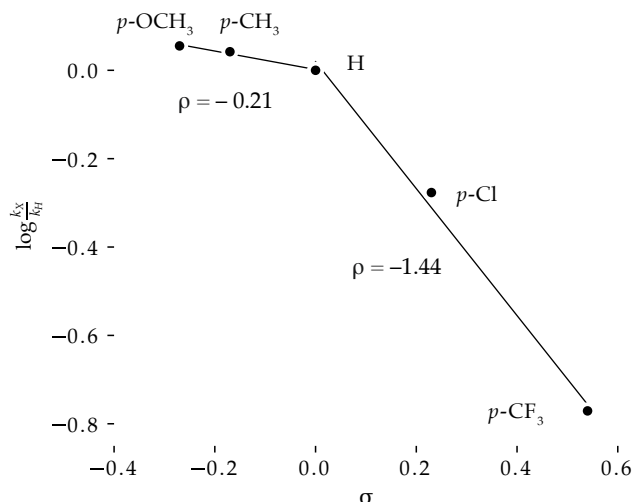
Table 4: Experimental results reported by the authors, calculations and substituent constants

Substituent	rel. rate	$\frac{k_X}{k_H}$	$\log \frac{k_X}{k_H}$
p-CF ₃	1.00	0.170	-0.771
p-Cl	3.12	0.529	-0.277
H	5.90	1	0
p-CH ₃	6.50	1.102	0.042
p-OCH ₃	6.70	1.136	0.055

⁸ In the series of compounds used to measure the *para* to *ortho* ratio presented in table 3, there is a sixth possible compound that the authors could have used in the Hammett series. Why is the relative rate for $R = -\text{OBz}$ not reported? Feel free to speculate.

Table 5: Experimental results reported by the authors, calculations and substituent constants

Substituent	σ	σ^+	σ^-
p-CF ₃	0.53		0.65
p-Cl	0.23	0.11	0.19
H	0		
p-CH ₃	-0.17	-0.31	
p-OCH ₃	-0.27	-0.78	



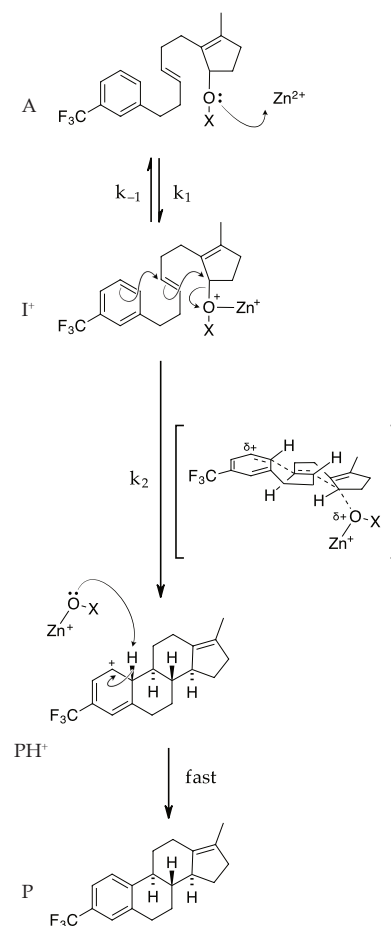
indicating a single mechanism with a change in the r.d.s. Starting at the right-hand side of the plot we see the ring-closing reaction becoming faster as the benzene ring becomes richer in electron density. Then we see the rate become almost independent of the benzene ring substituent with more electron-donating substituents. The ring-closing step involves the benzene ring so the step that is independent of substituent effects must be some other rate-limiting step.

The authors propose that the association of the Lewis acid with the leaving group (or the the formation of some type of activated complex involving the leaving group and the Lewis acid) becomes rate-limiting as the ring-closing reaction becomes faster. One might conclude that the region that is independent of the substituent represents the formation of an initial cation when the leaving group departs. This may well be true, but the leaving group seems to influence how the benzene ring adds to the alkene in the ring-closing process. If this is the case, there may be a tight ion-pair that separates only after the ring closing reaction. Either way, we clearly see a region that reveals that there is a substituent effect for the benzene ring being involved in ring closing. In figure 6, the full mechanism proposed by the authors is presented.

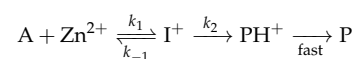
The formation of an electrophilic addition intermediate (PH^+) results in a full positive charge within the benzene ring itself. Reaction constants for such reactions typically are large. For example, the bromination of substituted benzenes has $\rho = -7.0$. The ρ value in this ring-closing reaction is much less. We conclude that the bond formation at the transition state is not very extensive at the benzene ring. The extent of bond formation in the two forming bonds may not be identical at the transition state.

Figure 5: The Hammett plot for the ring-closing reaction.

Figure 6: The proposed mechanism for the ring-closing reaction.



This scheme can be expressed as the following chemical equation...



and the rate law can be expressed as...

$$v = \frac{k_1 k_2}{k_{-1} + k_2} [\text{Zn}^{2+}] [\text{A}]$$

The concentration of zinc chloride is high and will not change significantly during the reaction (pseudo first-order conditions) so we could express the rate law as...

$$v = k_{\text{obs}} [\text{A}]$$

where...

$$k_{\text{obs}} = \frac{k_1 k_2}{k_{-1} + k_2} [\text{Zn}^{2+}]$$

Analysis of the Results

A More O’Ferrall-Jencks diagram may provide some insight. We will define the two forming bonds as the two axes. For the sake of this analysis, we will assume that the leaving group is in an ion-pair intermediate. It has departed but remains associated with the molecule. The positive charge on the carbocation of the activated intermediate will be transferred through the alkene to the benzene ring as the bonds form.

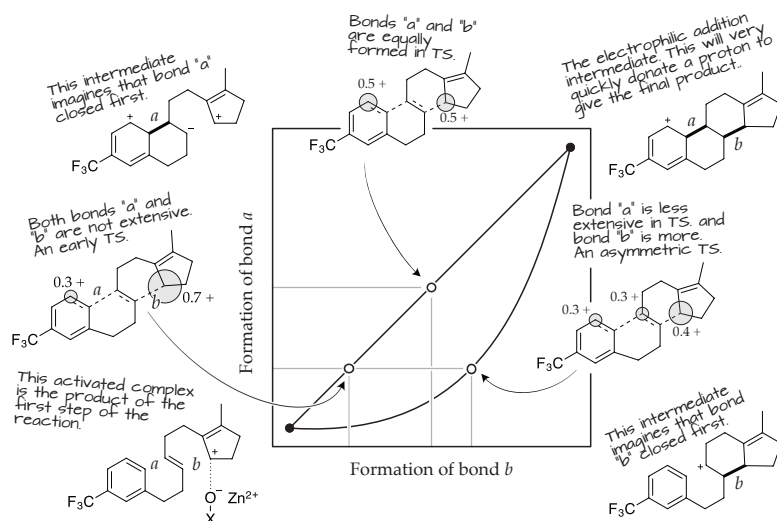


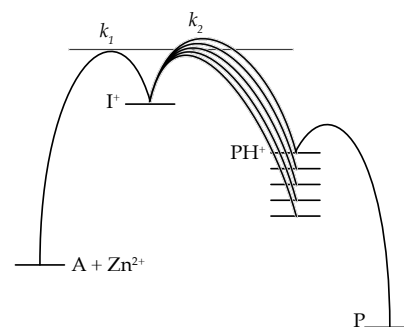
Figure 7: A More O’Ferrall-Jencks for the ring-closing reaction. Two possible routes and three potential transition states are described. Both the asymmetrical TS and the early TS models are consistent with the low magnitude of the ρ value. Which path do you favour?

We see that an early transition state can have the electron donation from the benzene ring towards the bonds beginning but not yet at a great extent. An asymmetric transition state where the alkene has donated more electrons to the forming bonds than the benzene would also show a similar reduced contribution for the aromatic electron pool. In both of these cases a reduced magnitude for the ρ value would be expected.

The reaction coordinate for the whole process is presented in figure 8. The formation of the initial activated complex is followed by the rapid ring-closing reaction. The fact that this reaction is so fast at cold temperatures is because it is an intramolecular process and the concerted mechanism provides a route to product that allows for the sharing of the charge among several carbons in the transition state. Take note of how the reaction coordinate changes in this diagram as the electron density ring is changed. The addition product, PH^+ , is most affected.

We were able to observe the reaction constant, ρ , for the ring-closing step, k_2 , and observed a significantly negative but smaller-than-expected value. We conclude that the ring-closing process is a concerted process with both bonds forming and the leaving group departure involved. The extent of bond formation at the benzene ring is small in the TS.

Figure 8: A reaction coordinate diagram for the scheme is figure 6



The k_2 reaction is rate limiting when the benzene ring is electron-deficient. With increasing electron density in the aromatic ring, the k_2 reaction becomes faster and the k_1 reaction becomes the r.d.s. The k_1 step does not involve the benzene ring and will not demonstrate substituent effects.

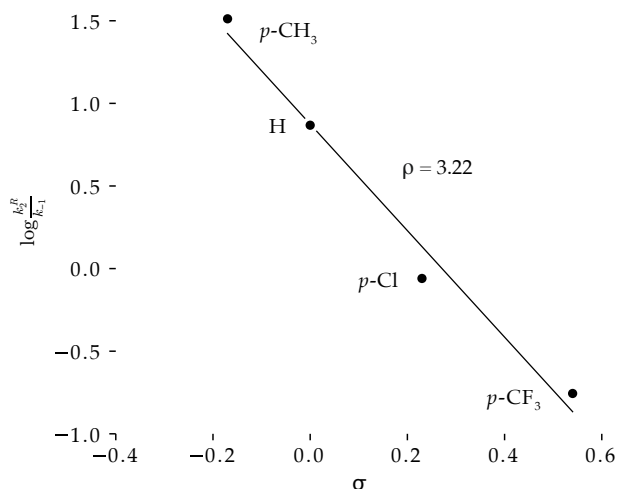
That Ratio Thing

The authors demonstrated a substituent effect on k_{obs} of $\rho = -1.4$. This is significant but is nowhere near what is expected for a reaction where the product has a charge residing within the former aromatic ring. They addressed this issue with math. Do you agree with their approach?

The rate constant for the ring-closing step, k_2 , is only part of the rate law. the authors state that a more rigorous approach to evaluating the effect of substituents on the rate constant for ring closing, k_2 , is to use the full rate equation. In pseudo first-order conditions, the observed rate constant is...

$$k_{obs} = \frac{k_1 k_2}{k_{-1} + k_2}$$

First they note that when the substituent is $-OCH_3$, we are at the extreme side of the region where k_1 is the rds. This means that $k_{obs}^{OCH_3} \approx k_1$. So if we take the ratio of $k_{obs}^{OCH_3}/k_{obs}^R$ we will get a term that gives us the ratio of k_2/k_{-1} . It is reasoned that both k_1 and k_{-1} do not change across the series as they are involved with the leaving group, not the substituted aromatic ring. So the k_2/k_{-1} ratio should reflect the change in the value of k_2 . The math is described in figure 9



The authors did not use the $p-CH_3$ data point in their treatment of the system and presented a slope or $\rho = -3$ for a three-point plot. Their math should work for every point (although we cannot use the $p-OCH_3$ point because we are using that value to represent k_1 .) With this more inclusive approach, we see a reaction constant of -3.2 , which is consistent with the authors' value. This still represents a smaller contribution of bonding from the aromatic ring in the TS, but it is a more reasonable value for that case.

Figure 9: Derivation of a relationship between $k_{obs}^{OCH_3}/k_{obs}^R$ and k_2^R .

Let us state that...

$$k_{obs}^{OCH_3} \approx k_1$$

and...

$$k_{obs}^R = \frac{k_1 k_2^R}{k_{-1} + k_2^R}$$

so...

$$\frac{k_{obs}^{OCH_3}}{k_{obs}^R} = \frac{k_1}{\frac{k_1 k_2^R}{k_{-1} + k_2^R}} = \frac{k_{-1} + k_2^R}{k_2^R} = \frac{k_{-1}}{k_2^R} + 1$$

We could get a value for k_2/k_{-1} as follows...

$$\frac{k_2^R}{k_{-1}} = \frac{1}{\frac{k_{obs}^{OCH_3}}{k_{obs}^R} - 1}$$

Figure 10: The Hammett plot for the ring-closing reaction using the calculated k_2^R/k_{-1} value to assess the substituent effect on the ring-closing step. The value for $k_{obs}^{OCH_3}$ was not included because we had used it as the approximation for the value of k_1 .

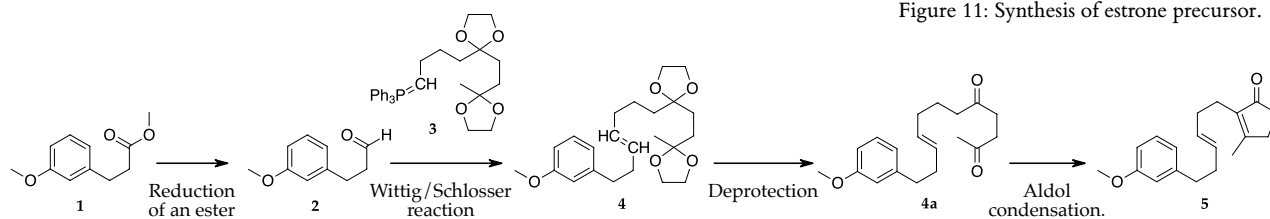
Table 6: Calculated values for k_2^R/k_{-1} according to the method used by the authors.

Substituent	rel. rate	k_2^R/k_{-1}	$\log(k_2^R/k_{-1})$
p-CF ₃	1.00	0.175	-0.756
p-Cl	3.12	0.872	-0.060
H	5.90	7.38	0.868
p-CH ₃	6.50	32.5	1.52
p-OCH ₃	6.70		

A Review of Advanced Organic Chemistry

The synthesis of the precursor for the ring-closing step involves a short series of reactions⁶ that you will be familiar with from your previous courses in organic chemistry. Some acronyms may be different but the principles are all the same.

The synthetic scheme for the synthesis of the reactant of the ring-closing reaction is presented in figure 11.



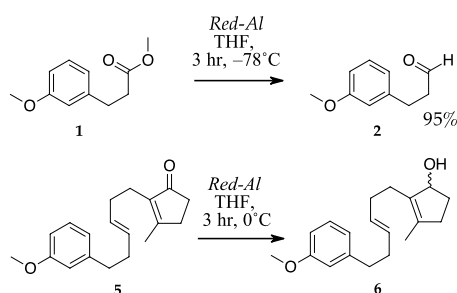
There are four steps for which you should be imagine reaction conditions and propose mechanisms. Read the paper¹ and observe the reaction conditions used by the authors. How do they compare to your own ideas. There are many ways to skin a cat so you may well have proposed excellent ideas that will work. The authors used their favourite methods and we will examine their choices.

The four steps are...

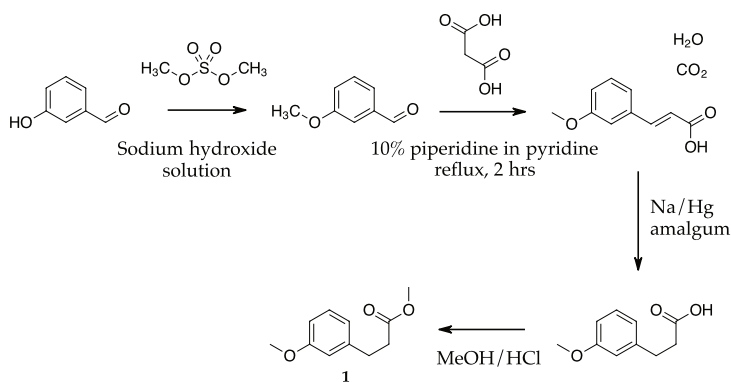
1. A hydride reduction of an ester. You will be familiar with LiAlH_4 (LAH) reducing esters to alcohols. The authors used a milder, safer reagent and conditions that favoured the production of the aldehyde rather than an alcohol. Be sure to examine this step. You will come away with a tool to reduce esters to aldehydes. Every synthesis is a lesson and you will always see new reagents or methods to do a familiar transformation.
2. A Wittig reaction. Here we will see a method that favours the thermodynamic *trans* product rather than the usual kinetically favoured *cis* product that you are familiar with in Wittig reactions.
3. Acid hydrolysis of a cyclic acetal. This is a simple reaction but we will use it as a starting point for discussing the idea of protecting carbonyl groups in synthetic strategies.
4. An aldol condensation. This step is a classic.
5. Did I say four? There is a final reduction reaction not shown in the scheme above. We will need to reduce that ketone to an alcohol to obtain the molecule 6 used in the physical organic studies that we explored in the first part of this document.

A mild reduction method

Let us begin our exploration of these five steps with the reduction reactions used at the beginning and end of the synthesis of **6**. In this work, the reducing agent "*Red-Al*" is used. It is a stabilized version of LiAlH_4 . The structure is shown in figure 12. The glycol groups stabilize the molecule and render it soluble in organic solvents. *Red-Al* is available as a solution in toluene.⁹ You can transfer it by syringe under nitrogen. You will recall that LiAlH_4 is a powder that bursts into flame in contact with water and it will produce only alcohols from esters.

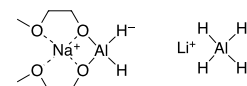


You will be familiar with DIBAL as a reducing agent for converting esters to aldehydes. When *Red-Al* is used, there is usually a cyclic amine added, such as N-methylpiperazine, to produce a more selective reducing agent.¹⁰ The authors did not report using piperazine, but I expect that they did.



The starting material, **1**, was synthesized according to previous work. From *m*-hydroxybenzaldehyde. The hydroxyl was methylated using dimethylsulfate. Then a condensation with malonic acid produced *m*-methoxycinnamic acid.¹¹ The double-bond of the cinnamic acid was then reduced with a sodium amalgam¹² and the methyl ester is made via con-

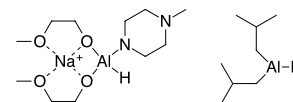
Figure 12: *Red-Al* and LiAlH_4



⁹ <https://www.sigmaaldrich.com/CA/en/product/aldrich/196193>

Figure 13: *Red-Al* reductions described in the synthesis. Observe that *Red-Al* will reduce a ketone to an alcohol so care must be taken that it does not do the same to the aldehyde **2**. Take note of the temperature difference.

Figure 14: Piperazine modified *Red-Al* and DIBAL are both useful for reducing esters to aldehydes.



¹⁰ "Partial Reduction of Esters to Aldehydes Using a Novel Modified Red-Al Reducing Agent", W.K. Shin, D. Kang, D.K. An., *Bull. Korean Chem. Soc.*, 2014, 35, 2169–2171. <https://doi.org/10.5012/bkcs.2014.35.7.2169>

Figure 15: Literature synthesis of **1**. Propose a different route using reactions that you are familiar with from your organic chemistry experience.

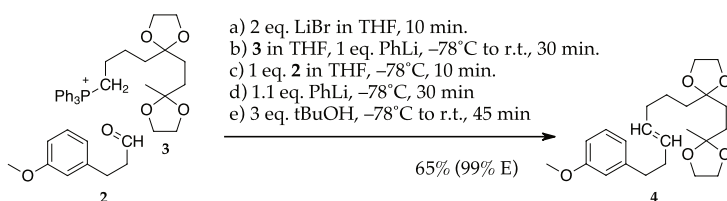
¹¹ "Über die katalytische Reduktion von as-Diaryltrichloräthanen," K. Brand, O. Horn, *Journal für Praktische Chemie*, 1927, 111, 351–378. <https://doi.org/10.1002/prac.19271150129>

¹² In the literature method, 1.2 kg of amalgam is used to reduce 55 g of the alkene. It is almost 100 years later. Today we would prefer to use catalytic hydrogenation

densation in acidic methanol.¹³ The synthesis is outlined in figure 15. The propionic acid precursor to 1 is now available from Sigma-Aldrich.¹⁴ Time is more valuable than money.

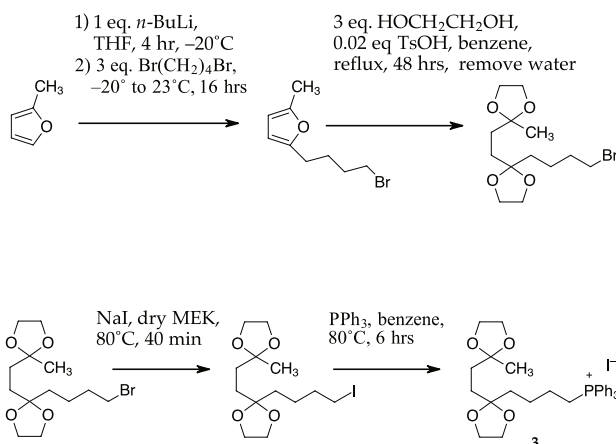
A Stereocontrolled Wittig Reaction

The next step in the synthesis is a Wittig reaction. Normally, Wittig reactions favour products with *Z* stereochemistry. The authors used the famous Schlosser modification to produce almost exclusively the *E* stereoisomer.



Review the Wittig reaction and try to understand each step. The Schlosser method is about favouring the thermodynamic product rather than the kinetic product, which is usually favoured due to rapid breakdown of the betaine intermediate to form the product.

The diacetal, 3, was synthesized from methylfuran in previous work from the same group.¹⁵



¹³ "The synthesis of compounds related to the sterols, bile acids, and oestrus-producing hormones. Part V. The synthesis of conjugated aryl-hexadienes, and their behaviour in the Diels-Alder reaction." A. Cohen, *J. Chem. Soc.*, 1935, 429-436. <https://doi.org/10.1039/JR9350000429>

¹⁴ <https://www.sigmaaldrich.com/CA/en/product/aldrich/349763>

Figure 16: Wittig reaction using the Schlosser modification. Steps a-c are the traditional Wittig reaction with LiBr added to stabilize the intermediate. Step d is the Schlosser step where we deprotonate the intermediate to allow the stereochemistry to equilibrate to the most stable conformer. In step e, the BuOH is an acid that neutralizes the intermediate and it collapses to product.

¹⁵ "Acetylenic bond participation in biogenetic-like olefinic cyclizations. II. Synthesis of dl-progesterone." W.S. Johnson, M.B. Gravestock, B.E. McCarry, *J. Am. Chem. Soc.*, 1971, 93, 4332-4334. <https://doi.org/10.1021/ja00746a062>.

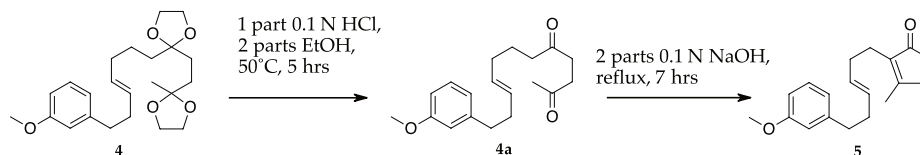
Figure 17: Synthesis of the diacetal from methylfuran. The first step features creation of a carbon nucleophile on furan via deprotonation (lithiation of furan) followed by an $\text{S}_{\text{N}}2$ reaction with an alkyl bromide. The second step is acid catalyzed addition to the furan to create the acetals.

Figure 18: Synthesis of the Wittig reagent. Iodide displaces bromide and then triphenylphosphine replaces the iodide.

The final two steps

The final two steps are an acid-catalyzed hydrolysis of the acetals to unmask the ketone groups that they are protecting. This is followed by an intramolecular aldol condensation.

Figure 19: Deprotection of the carbonyl groups followed by aldol condensation.



You should review every reaction presented here and understand the stepwise mechanism of each. You have seen all of these before.¹⁶

¹⁶ Where appropriate, propose a modernization of the methods. We no longer use benzene as a solvent and there are many ways to reduce an alkene other than using a pound of mercury.

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will...

- have explored more complex reaction schemes and observe how LFRE plots can curve depending on the changing relative rates of the steps or pathways.
- have gained more experience interpreting literature examples and following descriptions of chemical synthesis.
- **Computer Skill:** have again used an interactive *Python* notebook to interpret and analyze experimental data from the literature.